Example 7.22

Evaluate the following definite integrals:

(i) $\int_1^3 x^3\,dx$ (ii) $\int_{-1}^{+1} 5x^4\,dx$ (iii) $\int_1^2 \frac{5}{x}\,dx$

Solutions:

(i) $\int_1^3 x^3\,dx = \left[\frac{x^4}{4}\right]_1^3$

$= \frac{1}{4}[3^4 - 1^4]$

$= \frac{1}{4}[81 - 1]$

$= \frac{1}{4}[80]$

$= 20$

(ii) $\int_{-1}^{+1} 5x^4\,dx = 5\int_{-1}^{1} x^4\,dx$

$= 5\left[\frac{x^5}{5}\right]_{-1}^{1}$

$= [x^5]_{-1}^{1}$

$= [1^5 - (-1)^5]$

$= 1-(-1)$

$= 1+1$

$= 2$

(iii) $\int_1^2 \frac{5}{x}\,dx$

$= 5\int_1^2 \frac{1}{x}\,dx$

$= 5\,[\log x]_1^2$

$= 5[log2 - log1]$

$= 5log2$

7.5.3 Double integrals

A double integral is an integral of two variable function $f(x,y)$ over a region R If $R=[a, b] \times [c, d]$ then the double integral can be done by iterated Integration(integrate first with respect to y and then with respect to x)